MIL-H-51291A(EA)
22 February 1977
SUPERSEDING
MIL-H-51291(MU)
21 December 1967

MILITARY SPECIFICATION

HOOD, CHEMICAL-BIOLOGICAL MASK: M6A2

This specification is approved for use by Edgewood Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of protective hood.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

U-T-30 - Talcum Powder.
MMM-A-1617 - Adhesive, Rubber Base, General Purpose.

MILITARY

MIL-C-51251 - Cloth, Coated; CBR Protective.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Edgewood Arsenal, Attn: SAREA-TS-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.
FED-STD-175 - Adhesives: Methods of Testing.
FED-STD-191 - Textile Test Methods.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

D5-35-86 - Hood, Chemical-Biological Mask: M6A2
C18-74-322 - Air Permeability Test Fixture

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D5-35-86 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the hood shall conform to the specifications listed on Drawing D5-35-86 and on subsidiary drawings.

3.1.2.1 Slide fastener air permeability. The air permeability of the slide fastener shall not exceed 15 litres of air per minute when tested as specified in 4.4.4.5.

3.1.2.2 Slide fastener low temperature operation. The force required to operate the dry slide fastener, after conditioning, shall not exceed 2.0 pounds when tested as specified in 4.4.4.6.

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3.2 Assembly. The hood shall be assembled in accordance with Drawing D5-35-86.

3.2.1 Stitching. All rows of stitching shall be regular lines free from loose stitching or thread loops. All stitching shall be of continuous thread. All seams shall be backstitched at the ends with a minimum of 3 stitches. All loose thread ends shall be trimmed to 1/4 inch or less.

3.3 Fastening devices. The pile and hook fastener, drawcord, buckle, zipper, drawcord clip, and spring, shall function smoothly and efficiently.

3.4 Light penetration. When the hood is examined over a 100-watt light as specified in 4.4.4.1, there shall be no light penetration through any hole in the body or through six or more holes in the seams of the hood. However, the seams around the five openings in the front, along the zipper, and the area within 2.0 inches of the bottom of the hood shall be exempt.

3.4.1 Windows (film transparencies). Windows are permissible. However, windows or light areas that are suspected of containing a hole shall be tested as specified in 4.4.4.1.

3.4.2 Luting compound. A luting compound may be used on the seams of the hood prior to the light penetration test. The luting compound may be applied on the inside or the outside of the hood. When used on the outside, the luting compound shall be applied neatly and shall match the color of the hood. The seams shall be treated with an approved preparation (see 6.4) prior to the application of the luting compound. The luting compound shall be a self-curing solvent neoprene rubber diisocyanate adhesive (see 6.3). When used in combination with the coated fabric, the luting compound shall meet the following requirements when tested as specified in 4.4.4.2.

Blocking of seams	Scale rating No. 3
Resistance to low temperature	Pass test with no cracking, flaking, or separation from the fabric
Stiffness (centimetres)	
Initial	10 max
*After heat aging	12 max
Solids content	20 min

*The specimen shall show no cracking, flaking, or separation from the fabric.

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3.4.3 Dusting. After the luted seams have thoroughly dried, they shall be dusted with talcum powder conforming to U-T-30.

3.5 Fit. The hood must assemble to the M17 and M17A1 masks without distortion of the hood or the mask (see 4.4.4.3).

3.6 Repairs. Repairs are permitted on the hood for cuts, nicks (1/4-inch maximum length), or pinhole punctures. Repairs shall be accomplished by securely cementing an approximate 1 1/2-inch diameter patch of material conforming to MIL-C-51251 to the inside of the hood so as to completely cover the defective area. The cement shall conform to type I or type II of MMM-A-1617. Prior to the application of the cement, the surface shall be prepared with Hylene M-50 (see 6.4). No more than one repair will be permitted on any single hood and each repaired hood shall be considered a minor defect when examined in accordance with 4.4.3.3.

3.7 Preproduction. Prior to the start of regular production, a preproduction sample shall be produced, examined, and tested in accordance with this specification (see 4.3).

3.8 Color. The coated-cloth components of the finished hood shall be a commercial match to color number 34087 of FED-STD-595; or, alternatively, the color and shade of the finished hood shall be OD 177. In either case, the color shall be tested as specified in 4.4.4.7.

3.9 Workmanship. The hood shall be free from damage (tear, rip, cut, nick, abrasion, or puncture) and foreign matter (dirt, stain, grease, or oil). All seams shall be free from pleating or distortion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classsification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 100 hoods and one sample seam (4.4.4.2.1) shall be produced using the same methods, materials, equipment, and processes as will be used in regular production. Changes in manufacturing techniques will require a new preproduction sample.

4.3.2 Examination and tests. The entire preproduction sample shall be examined and tested for all the requirements of the drawings and of this specification except low temperature operation of the slide fastener (4.4.4.6) shall be conducted on five hoods selected at random from the preproduction sample.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the hoods produced by one manufacturer, at one plant, from the same materials using one color of coated cloth, and under essentially the same manufacturing conditions. However, no more than five consecutive lots of butyl coated cloth from one manufacturer shall be represented in any one lot of finished hoods; the strap, binder strip, reinforcing strips, and reinforcing patch components need not be considered in the application of this lotting criteria.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105. For major defect 123 level S-1 shall be used.

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4.4.2.2 For pattern configuration. For die-cut blanks, sampling shall be conducted in accordance with MIL-STD-105, level S-1. When blanks are knife-cut, two samples shall be taken from each blank cutting, one from the top and one from the bottom.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. Sample hoods and the level A preparation for delivery thereof shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For pattern configuration. Sample blanks shall be examined in compliance with the applicable pattern. The acceptance number is zero.

4.4.3.3 Classification of defects.

(a) Hood, chemical-biological mask: M6A2 (Dwg D5-35-86).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.65 percent defective	
101	Light penetration through hood	4.4.4.1
102	Light penetration through six or more needle holes in seams (see 3.4)	
	AQL 1.0 percent defective	
103	Heavy coated side of cloth not on exterior of hood	
104	Component match marks misaligned by more than 1/8 inch	
105	Distortion of pleating affecting fit	
106	Component missing or incorrect	
107	Component incorrectly assembled	
108	Fastening devices (pile and hook fastener, drawcord, buckle, zipper, drawcord clip, or spring) not functioning correctly	

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109	Stitching run off or stitches broken more than 1/2-inch	
110	Number of stitches per inch incorrect (two more or less than specified)	
111	Dusting of luted seams with talcum powder insufficient to prevent tackiness	
112	Marking incorrect	
113	Damage (tear, rip, cut, nick, puncture, or abrasion) or foreign matter other than 208	
114	Stitching margins not as specified	
115	Misalignment of inner lining and hood body by more than 1/8-inch	
116	Overlap of reinforcement strip less than 3/8-inch	
117	Dimensions of inlet valve or lens opening not as specified	4.4.4.4
118	Separation of coating	
119	Brittle areas which crack when bent	
120	Blisters or lumps	
121	Any area of no coating	
122	Colors and shades not as specified	4.4.4.7
	AQL 2.5 percent defective	
123	Fit	4.4.4.3
Minor:	AQL 6.5 percent defective	
201	Distortion or pleating not affecting fit	
202	Stitching run off or broken more than 1/4-inch but less than 1/2-inch	
203	Stitching tension loose, causing exposed loops, or tight, causing pleating	
204	Thread ends not trimmed to 1/4-inch or less	
205	Number of stitches per inch incorrect (one more or less than specified)	

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206	Seam not backstitched at least three stitches (unless held by another row of stitches)
207	Stitch type or pattern not as specified
208	Minor foreign matter not affecting serviceability (small areas of stain, dirt, or machine lubricant, other than oil or grease not exceeding 1/4-square inch area and a total area of foreign matter not exceeding 1 square inch)
209	Marking illegible or improperly located
210	Repaired hoods (3.6)
211	Overlap of reinforcement strip more than 3/8-inch but less than 1/2-inch

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
*101	Hoods not dusted with talcum powder
*102	Hood color and shade incorrect (3.8)

*Check prior to sealing.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Light penetration. The hood shall be examined over the specified light at rated voltage for light penetration. The light shall be no farther than 12 inches from the area examined and the observer no farther than 24 inches. Each seam shall be examined from at least two angles. The light intensity of the area around the hood being examined shall be no more than 32 footcandles. Windows or light areas which are suspected of containing holes shall be subjected to a hydrostatic pressure of two pounds per square inch minimum for 60 ± 5 seconds and shall show no evidence of water leakage.

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4.4.4.2 Luting compound.

4.4.4.2.1 Samples. Approximately 2 yards of sample seam, 6 inches wide, shall be prepared by joining two pieces of the coated cloth with the type of stitch and seam defined in Section H-H on Drawing D5-35-86. The seam shall be coated with luting compound after it has been treated with an approved preparation (see 6.4).

4.4.4.2.2 Blocking resistance. Resistance to blocking of the luting compound shall be tested on a sample prepared as specified in 4.4.4.2.1. Test method 5872 of FED-STD-191 shall be used with the following exceptions:

(1) Test specimens shall measure approximately 6 x 8 inches with the seam in the center running the full 8-inch length.

(2) The seam shall be folded over on itself once to a size of approximately 6 x 4 inches so that the luting compound is on the outside of the folded specimen.

(3) The test temperature shall be 71.1 ± 1.0° C.

(4) A 1-pound weight shall be distributed evenly along the seam.

4.4.4.2.3 Low temperature resistance. The resistance of the luting compound to low temperatures shall be tested on a sample prepared as specified in 4.4.4.2.1. Test method 5874 of FED-STD-191 shall be used with the following exceptions:

(1) Test specimens shall measure approximately 6 by 8 inches with the seam in the center running the full 8-inch length.

(2) The seam shall be folded over on itself to a size of approximately 6 by 4 inches so that the luting compound is on the inside of the folded specimen.

(3) The roller weight shall be run along the fold line.

4.4.4.2.4 Stiffness.

4.4.4.2.4.1 Initial. A two-mil film (dry thickness) of the luting compound, approximately 3 by 12 inches, shall be cast on the coated cloth. The film shall be allowed to air dry for 24 hours. Test

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specimens shall be cut from the above prepared cloth and the stiffness shall be determined as specified in method 5204 of FED-STD-191.

4.4.4.2.4.2 After heat aging. Test specimens shall be prepared as specified in 4.4.4.2.4.1 and then placed in a circulating air oven at 82.2 ± 1.0° C for 24 hours. After the specimens are removed from the oven, they shall be allowed to cool for a minimum of 1 hour and then the stiffness determined as specified in method 5204 of FED-STD-191.

4.4.4.2.5 Solids content. The solids content of the luting compound shall be determined as specified in method 4021 of FED-STD-175.

4.4.4.3 Fit (see 6.5).

4.4.4.3.1 Distortion. The hood shall be assembled to an M17 and M17A1 mask and observed for stressed seams in the hood and for distortion of the hood or mask. (Stretching of the inlet valve or the lens opening of the hood to effect assembly to the mask is not considered distortion nor is puckering or pleating of the hood material if it does not result in stressing of the seams.)

4.4.4.3.2 Voicemitter opening spring.

4.4.4.3.2.1 Outlet valve inside hood. The hood shall be assembled to an M17 and an M17A1 mask with the outlet valve inside the hood. The spring in the voicemitter opening must not cover any portion of the holes in the round guard of the diaphragm in the voicemitter.

4.4.4.3.2.2 Outlet valve outside hood. The hood shall be assembled to an M17 and an M17A1 mask with the outlet valve outside the hood. The spring in the voicemitter opening must fit the underside of the voicemitter outlet valve assembly snugly.

4.4.4.4 Inlet valve and lens openings. The inlet valve and lens openings shall be assembled on a split-plug fixture which approximates the configuration of the lens or inlet valve openings as applicable and applies a circumferential tension of 25.0 ± 1.0 pounds to the inlet valve opening or 20.0 ± 1.0 pounds to the lens opening. The inside circumference of the openings shall be determined under these tensions and shall conform to Drawing D5-35-86.

4.4.4.5 Slide fastener air permeability. The air permeability of the slide fastener shall be determined using method 5450 of FED-STD-191 except that the slide fastener shall be assembled in a test fixture as shown on Drawing C18-74-322.

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4.4.4.6 Slide fastener low temperature operation. The slide fastener shall be conditioned at -54° C for four hours and then while closed shall be laid flat, without tension, on a horizontal surface with both ends fastened down. The chain shall be opened and closed three times, over at least a 5-inch portion, by drawing the slider back and forth, as normally operated, by means of a calibrated spring scale attached to the slider pull. A maximum load indicator shall be attached to the spring scale and the load indicated for each opening and closing cycle shall be recorded.

4.4.4.7 Color. The color of the coated-cloth components shall be determined in accordance with method 4250 of FED-STD-141. Before making color matchings the dusting powder shall be removed from the hood by wiping with a damp cloth.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Hoods shall be prepared for delivery as specified on the packaging data sheet which is identified by the National Stock Number for the hood. Packaging shall be level A, and packing shall be level A or Commercial as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The M6A2 hood is intended for use within the M17 and M17A1 chemical-biological masks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Packing level (A or Commercial).
(c) Preproduction

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility when testing is performed by the Government.
(3) Packing and shipping instructions when applicable.
(4) Time required for the Government to notify the supplier whether or not to proceed with production.

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6.3 Luting compound. The following compound has been used satisfactorily and may be used provided it meets the requirements of 3.4.2.

Part I	*Part II
100.0 parts by weight of Neoprene (type WRT) 4.0 parts by weight of calcined MgO (extra light) 2.0 parts by weight of Neozone D 5.0 parts by weight of zinc oxide 145.0 parts by weight of methylethyl-ketone 75.0 parts by weight of Toluene 2.5 parts by weight of Thiuram E plus dye	10 parts by weight of Hylene M-50 *Shall be added to Part I just prior to use

6.4 Hylene M-50. Hylene M-50 has been used satisfactorily to prepare the seams for the luting compound. Hylene M-50, an organic isocyanate, is obtainable from E. I. DuPont de Nemours & Co., Inc., 1007 Market Street, Wilmington, Delaware.

6.5 Fit. The M17 and M17A1 masks required for use in performing the fit test will be furnished by the Government and should be obtained as directed by the contracting officer.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A528

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau N[illegible]25

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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